

# Expanded Cybersecurity Internship Documentation

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## Introduction

This document contains structured cybersecurity internship notes organized into major concepts, networking fundamentals, practical tools, scanning, enumeration, password security, and virtual lab environments. The document is designed as beginner-friendly crash course material for revision, understanding, and practical learning.

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## Month 1 — Cybersecurity Foundations & Networking

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### 1. Cybersecurity Basics

#### What is Cybersecurity?

Cybersecurity is the practice of protecting systems, networks, servers, applications, and data from cyberattacks, unauthorized access, malware, and digital threats.

Cybersecurity focuses on:

- Confidentiality
- Integrity
- Availability

These three together are often called the CIA Triad.

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#### Why Cybersecurity is Important

Cybersecurity is important because modern organizations, governments, and individuals rely heavily on digital systems.

Without cybersecurity:

- Data can be stolen
- Systems can be damaged
- Financial losses can occur
- Privacy can be violated
- Services can become unavailable

## **Practical Importance**

Cybersecurity helps:

- Protect sensitive information
  - Prevent cyberattacks
  - Secure online communication
  - Maintain business continuity
  - Protect users and organizations
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## **2. Importance of Cybersecurity in AI Era**

Artificial Intelligence systems process large amounts of sensitive and important data.

As AI becomes more common in:

- healthcare
- banking
- automation
- education
- defense

cybersecurity becomes even more important.

## **Why It Matters**

### **Data Protection**

AI systems rely on large datasets. If attackers steal or manipulate data, AI systems may behave incorrectly.

### **Model Manipulation**

Attackers may try to manipulate AI models using malicious inputs.

### **Privacy Risks**

AI systems often handle personal and sensitive user information.

## **Automation Risks**

Compromised AI systems can automate attacks or make incorrect decisions.

## **Practical Usage**

Cybersecurity in AI is used for:

- securing AI applications
  - protecting datasets
  - monitoring threats
  - preventing unauthorized access
  - protecting automated systems
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# **3. Hacking vs Ethical Hacking**

## **Hacking**

Hacking refers to gaining access to systems, networks, or devices.

Malicious hacking is illegal and performed to:

- steal data
  - damage systems
  - disrupt services
  - gain unauthorized access
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## **Ethical Hacking**

Ethical hacking is legal security testing performed with permission.

Ethical hackers identify vulnerabilities before attackers exploit them.

## **Why Ethical Hacking Matters**

- improves security
  - identifies vulnerabilities
  - helps organizations strengthen defenses
  - prevents future attacks
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## 4. Types of Hackers

### White Hat Hackers

Ethical hackers who work legally to improve security.

#### Main Role

- penetration testing
  - vulnerability assessment
  - security auditing
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### Black Hat Hackers

Hackers who perform illegal activities for malicious purposes.

#### Main Activities

- stealing data
  - spreading malware
  - financial fraud
  - unauthorized access
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### Grey Hat Hackers

Hackers who may violate rules but do not always act maliciously.

They sometimes discover vulnerabilities without permission and later report them.

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### Script Kiddies

Beginners who use existing tools/scripts without deep technical understanding.

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### Hacktivists

Hackers motivated by political or social causes.

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## State-Sponsored Hackers

Hackers supported by governments for espionage or cyber warfare.

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## 5. Phases of Hacking

### 1. Reconnaissance

Gathering information about the target.

### 2. Scanning

Identifying open ports, systems, and services.

### 3. Gaining Access

Accessing the target system.

### 4. Maintaining Access

Keeping continued access to the compromised system.

### 5. Covering Tracks

Removing evidence and hiding activities.

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## 6. Cyber Kill Chain

The Cyber Kill Chain is a cybersecurity framework developed by Lockheed Martin.

It explains the stages of a cyberattack.

### Phases

1. Reconnaissance
2. Weaponization
3. Delivery
4. Exploitation
5. Installation
6. Command & Control

## Defense Mechanisms Discussed

- Firewalls
  - Antivirus
  - IDS/IPS
  - Monitoring
  - Email filtering
  - Backups
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## 7. Malware

### What is Malware?

Malware stands for malicious software.

It is software designed to:

- damage systems
  - steal information
  - disrupt operations
  - gain unauthorized access
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## Types of Malware

### Virus

Malicious code that attaches itself to files or programs.

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### Worm

Self-replicating malware that spreads automatically across networks.

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### Trojan

Malware disguised as legitimate software.

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## **Ransomware**

Malware that encrypts files and demands payment.

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## **Spyware**

Malware that secretly monitors user activities.

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## **Adware**

Software that displays unwanted advertisements.

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## **Rootkit**

Malware designed to hide malicious activities and maintain privileged access.

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# **8. Networking Fundamentals**

## **What is a Network?**

A network is a collection of connected devices that communicate and share resources.

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## **What is Networking?**

Networking is the process of connecting systems and devices for communication.

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# **9. Types of Networks**

## **LAN**

Local Area Network.

## **MAN**

Metropolitan Area Network.

## **WAN**

Wide Area Network.

## **PAN**

Personal Area Network.

## **CAN**

Campus Area Network.

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# **10. Network Topologies**

## **Bus Topology**

All devices share one communication line.

## **Star Topology**

Devices connect to a central switch or hub.

## **Ring Topology**

Devices form a circular communication path.

## **Mesh Topology**

Devices connect directly to multiple devices.

## **Tree Topology**

Hierarchical network structure.

## **Hybrid Topology**

Combination of multiple topologies.

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# 11. Networking Devices

## Hub

Broadcasts data to all connected devices.

## Switch

Sends data only to intended devices.

## Router

Connects different networks.

## Firewall

Filters incoming and outgoing traffic.

## NIC

Network Interface Card connecting device to network.

## Load Balancer

Distributes traffic across systems.

## Wireless Router

Provides wireless connectivity.

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# 12. IP Addressing

## What is an IP Address?

An IP address is a unique identifier assigned to devices on a network.

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# Types of IP Addresses

## IPv4

32-bit addressing system.

Example: 192.168.1.1

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## IPv6

128-bit addressing system.

Example: 2001:db8::1

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## Public IP Address

Accessible through the internet.

Assigned by ISP.

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## Private IP Address

Used inside local/internal networks.

Common ranges:

- 192.168.x.x
  - 10.x.x.x
  - 172.16.x.x – 172.31.x.x
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## Static IP Address

Manually assigned fixed IP address.

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## Dynamic IP Address

Automatically assigned IP address.

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## Loopback Address

Used for testing the local machine.

Range: 127.0.0.0 – 127.255.255.255

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## 13. IP Classes

### Class A

Large networks.

### Class B

Medium networks.

### Class C

Small networks.

### Class D

Multicasting.

### Class E

Experimental purposes.

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## 14. Ports & Protocols

### What are Logical Ports?

Logical ports are communication endpoints used by services and applications.

Ports help systems identify which service should receive network traffic.

Total logical ports: 65,535

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## Port Ranges

| Port Range  | Type                  |
|-------------|-----------------------|
| 0-1023      | Well-known Ports      |
| 1024-49151  | Registered Ports      |
| 49152-65535 | Dynamic/Private Ports |

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## Important Ports Table

| Port | Protocol/Service | Purpose                  |
|------|------------------|--------------------------|
| 20   | FTP Data         | File transfer data       |
| 21   | FTP              | File transfer control    |
| 22   | SSH              | Secure remote login      |
| 23   | Telnet           | Insecure remote login    |
| 25   | SMTP             | Email sending            |
| 53   | DNS              | Domain resolution        |
| 80   | HTTP             | Web communication        |
| 110  | POP3             | Email retrieval          |
| 139  | NetBIOS          | Windows networking       |
| 143  | IMAP             | Email retrieval          |
| 161  | SNMP             | Network management       |
| 162  | SNMP Trap        | SNMP notifications       |
| 443  | HTTPS            | Secure web communication |
| 445  | SMB              | Windows file sharing     |
| 3306 | MySQL            | Database service         |
| 3389 | RDP              | Remote desktop           |
| 5432 | PostgreSQL       | Database service         |
| 5900 | VNC              | Remote desktop control   |
| 8080 | Alternate HTTP   | Alternative web service  |

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## 15. NAT & DMZ

### NAT

Network Address Translation converts private IP addresses into public IP addresses.

#### Why It Matters

- conserves IP addresses
- hides internal systems
- enables internet access

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### DMZ

Demilitarized Zone is a buffer network between the internet and internal network.

#### Why It Matters

Provides additional security for public-facing services.

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## 16. Servers

### Web Server

Hosts websites.

### DNS Server

Converts domains into IP addresses.

### DHCP Server

Automatically assigns IP addresses.

### FTP Server

Provides file transfer services.

## **Email Server**

Handles email communication.

## **Database Server**

Stores and manages databases.

## **Proxy Server**

Acts as intermediary between client and internet.

## **File Server**

Stores and shares files.

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# **17. Hosting Types**

## **Shared Hosting**

Multiple users share the same server.

## **Dedicated Hosting**

Single user receives full server resources.

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# **18. Reconnaissance**

Reconnaissance is the process of gathering information about a target.

## **Active Reconnaissance**

Direct interaction with the target.

## **Passive Reconnaissance**

Indirect information gathering.

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# Month 2 — Virtual Labs, Linux & Scanning

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## 19. Virtual Lab Setup

### VirtualBox Installation

#### What is VirtualBox?

VirtualBox is virtualization software used to create and run virtual machines.

#### Why It Matters

Used to create cybersecurity labs safely.

#### Basic Installation Steps

1. Download VirtualBox
2. Run installer
3. Complete installation wizard
4. Launch VirtualBox
5. Create virtual machine

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### Kali Linux Installation

#### What is Kali Linux?

Kali Linux is a penetration testing Linux distribution.

#### Why It Matters

Used for cybersecurity testing and ethical hacking.

#### Basic Installation Steps

1. Download Kali Linux ISO
2. Open VirtualBox
3. Create new VM
4. Select Linux OS type
5. Allocate RAM and storage
6. Attach Kali ISO
7. Start installation
8. Complete setup process

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## Metasploitable Installation

### What is Metasploitable?

Metasploitable is an intentionally vulnerable virtual machine.

### Why It Matters

Used for safe cybersecurity practice.

### Basic Setup Steps

1. Import/download VM
2. Configure networking
3. Start virtual machine
4. Connect Kali Linux to target VM

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## Networking Modes

### NAT

VM accesses internet through host.

### NAT Network

Allows VM communication inside NAT environment.

### Bridged Adapter

VM appears as real device on network.

### Internal Network

Only internal VM communication allowed.

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## 20. Kali Linux Basics

### Root User

Administrator account with highest privileges.

## Terminal Usage

Command-line interaction with Linux.

## Linux File Navigation

Managing directories and files.

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## Linux Commands

| Command  | Purpose                    |
|----------|----------------------------|
| pwd      | Show current directory     |
| whoami   | Show current user          |
| ip a     | Display IP addresses       |
| ifconfig | Display network interfaces |
| ping     | Test connectivity          |
| sudo su  | Switch to root user        |
| clear    | Clear terminal screen      |
| nano     | Open text editor           |
| ls       | List files/folders         |
| cd       | Change directory           |

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## 21. Host Discovery & Scanning

### Netdiscover

Discovers live hosts using ARP.

### Angry IP Scanner

Fast network scanning tool.

## Advanced IP Scanner

Windows-based network scanner.

## Ping Sweep

Technique for identifying live hosts.

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## 22. Nmap

Nmap is a network scanning and enumeration tool.

### Main Uses

- host discovery
  - port scanning
  - service detection
  - OS detection
  - enumeration
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## Important Nmap Commands

```
nmap target  
nmap -F target  
nmap -p- target  
nmap -sS target  
nmap -sU target  
nmap -O target  
nmap -sV target  
nmap -A target  
nmap -sC target
```

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## 23. Zenmap

Graphical user interface version of Nmap.

## Features

- topology visualization
  - scan profiles
  - result visualization
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# Month 3 — Enumeration, Passwords & Practical Labs

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## 24. Enumeration

Enumeration is the process of extracting detailed information from services.

### Types Discussed

- FTP Enumeration
  - SSH Enumeration
  - Telnet Enumeration
  - SSL/TLS Enumeration
  - SNMP Enumeration
  - Database Enumeration
- 

## 25. FTP

File Transfer Protocol used for transferring files.

### Ports

- 20
  - 21
- 

## 26. Telnet

Insecure remote login protocol.

## Port

23

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## 27. SSH

Secure remote login protocol.

## Port

22

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## 28. SNMP Enumeration

Used for network device information gathering.

## Ports

- 161
  - 162
- 

## 29. Database Enumeration

Interacting with databases such as MySQL/MariaDB.

## Port

3306

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## 30. Netcat

Tool used for:

- banner grabbing
  - manual interaction
  - shell interaction
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## 31. Password Cracking Basics

Password cracking involves recovering passwords from hashes.

### Main Concepts

- weak passwords
  - password hashes
  - offline cracking
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## 32. Ophcrack

Windows password cracking tool.

### Main Uses

- password recovery
  - LM/NTLM cracking
  - password auditing
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## 33. Rainbow Tables

Precomputed tables of password hashes.

### Why They Matter

Used to speed up password cracking.

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## 34. LM & NTLM Hashes

### LM Hash

Older and weak Windows hash format.

### NTLM Hash

Newer Windows hash format.

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## 35. Weak Password Security

Weak passwords are vulnerable to:

- brute force attacks
  - dictionary attacks
  - rainbow table attacks
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## Important Practical Skills Learned

- VM networking
- host discovery
- scanning
- enumeration
- service interaction
- SSL analysis
- database interaction
- vulnerability identification
- password auditing